

TACOMA NARROWS, WA, USA

WMO: 727938

Lat: 47.268N

Lon: 122.576W

Elev: 292

StdP: 14.54

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94274

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	27.7	31.0	13.6	11.1	33.2	19.3	14.7	35.5	24.9	46.7	22.1	46.6	5.2	10	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	83.8	64.3	80.6	63.0	76.9	61.8	65.8	80.7	64.2	77.4	62.8	74.5	7.1	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	80.3	68.3	59.1	75.6	67.1	57.8	72.0	65.9	30.6	80.8	29.4	77.4	28.4	74.6	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	17.2	15.3	DB	23.9	91.2	3.9	4.5	21.1	94.5	18.9	97.1	16.7	99.7	13.9	103.0	(4)
(5)			WB	22.1	68.6	4.3	1.6	19.0	69.8	16.5	70.7	14.1	71.6	11.0	72.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	42.0	42.9	46.0	49.9	55.8	60.4	65.2	65.2	60.9	52.7	45.7	41.3	(6)	
	DBStd	9.87	5.40	4.91	4.64	4.65	5.36	5.20	4.70	3.92	4.35	4.04	5.24	5.14	(7)	
	HDD50	1093	251	201	139	57	8	0	0	0	0	18	147	272	(8)	
	HDD65	4755	712	619	588	454	290	160	53	45	138	382	580	734	(9)	
	CDD50	1967	4	2	17	53	188	311	472	473	326	102	17	2	(10)	
	CDD65	152	0	0	0	0	5	21	60	52	14	0	0	0	(11)	
	CDH74	1394	0	0	0	6	64	214	569	432	109	0	0	0	(12)	
	CDH80	333	0	0	0	0	8	56	164	90	15	0	0	0	(13)	
(14) Wind	WSAvg	6.1	6.3	6.7	7.0	6.5	6.1	6.1	5.6	5.2	5.3	5.8	6.6	6.3	(14)	
(15) Precipitation	PrecAvg	48.7	7.2	5.0	5.2	3.4	2.2	1.6	0.7	1.1	1.5	5.0	7.7	7.8	(15)	
	PrecMax	59.7	14.6	12.6	10.6	7.7	4.5	3.3	1.9	3.4	5.0	11.0	18.3	13.1	(16)	
	PrecMin	35.3	3.4	0.2	1.1	0.9	0.2	0.2	0.2	0.0	0.1	0.8	1.7	2.8	(17)	
	PrecStd	7.8	2.7	2.6	2.2	1.9	1.1	0.8	0.5	1.0	1.2	2.5	3.7	3.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.6	57.2	66.6	73.0	80.5	86.9	90.3	87.5	82.1	69.6	60.8	56.1	(19)	
		MCWB	53.0	49.6	52.3	56.4	60.5	64.0	67.6	65.3	62.1	56.9	56.6	52.9	(20)	
	2%	DB	53.5	54.3	60.0	66.2	74.9	80.8	84.2	82.3	77.2	64.3	57.2	52.4	(21)	
		MCWB	50.1	48.6	50.0	52.9	58.2	62.5	65.1	64.1	61.7	55.7	54.1	49.8	(22)	
	5%	DB	51.3	52.0	55.8	62.0	70.4	75.1	80.3	79.1	73.1	61.8	54.9	50.1	(23)	
		MCWB	48.5	47.2	48.7	51.3	56.9	59.9	63.4	63.0	60.3	54.9	52.1	47.4	(24)	
	10%	DB	49.1	49.9	53.5	57.8	65.8	70.9	76.1	75.4	69.9	59.1	53.0	47.9	(25)	
		MCWB	46.7	46.0	47.4	49.1	55.0	58.4	61.9	61.7	58.9	53.8	50.5	45.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.4	53.7	54.7	57.7	62.2	65.9	69.3	67.2	65.2	60.8	58.0	53.8	(27)	
		MCDB	55.9	55.2	61.7	70.1	75.5	83.1	87.3	82.3	76.2	64.6	59.8	55.8	(28)	
	2%	WB	51.3	50.4	52.1	54.6	59.8	63.2	66.3	65.4	63.2	58.1	54.7	50.5	(29)	
		MCDB	52.3	52.5	57.1	63.8	72.6	78.4	81.8	79.6	73.3	61.8	56.6	52.1	(30)	
	5%	WB	48.9	48.2	50.3	52.5	57.8	61.0	64.3	64.0	61.7	56.3	52.9	47.8	(31)	
		MCDB	50.3	50.5	54.8	59.4	68.3	73.2	78.0	76.7	70.3	59.8	54.6	49.4	(32)	
	10%	WB	46.7	46.3	48.3	50.6	55.8	59.2	62.6	62.5	60.1	54.7	50.8	46.0	(33)	
		MCDB	48.3	48.7	52.5	56.4	64.5	69.6	74.5	73.7	67.5	58.0	52.5	47.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	9.6	11.9	13.8	15.8	16.6	19.3	18.5	15.6	10.9	8.7	7.2	(35)	
		MCDBR	9.8	12.9	16.5	20.5	24.1	24.4	26.1	24.7	21.8	15.2	9.9	8.7	(36)	
	5% WB	MCWBR	6.7	7.9	8.7	9.9	11.1	10.5	10.4	9.9	9.2	8.2	7.1	6.7	(37)	
		MCWBR	8.4	10.0	14.3	17.6	20.8	21.8	23.5	22.0	18.3	12.1	8.9	8.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.327	0.314	0.327	0.353	0.375	0.363	0.349	0.354	0.342	0.329	0.324	0.331	(40)	
		taud	2.481	2.554	2.532	2.431	2.389	2.445	2.507	2.498	2.526	2.566	2.521	2.403	(41)	
	Ebn at Noon		240	274	286	286	282	285	288	283	277	264	240	220	(42)	
		Edh at Noon	21	23	28	34	36	35	32	31	28	23	20	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		322	594	886	1297	1643	1796	2006	1716	1258	711	378	265	(44)	
	RadStd		48	81	106	123	137	185	154	119	117	79	50	39	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page